

**Medicinal Actions & Uses** Henbane is used extensively in herbal medicine as a sedative and painkiller. Its specific use is for pain affecting the urinary tract, especially pain due to kidney stones, though it is also given for abdominal cramping. Its sedative and antispasmodic effect makes it a valuable treatment for the symptoms of Parkinson's disease, relieving tremor and rigidity during the early stages of the illness. Henbane has also been used to treat asthma and bronchitis, usually as a "burning powder" or in the form of a cigarette. Applied externally as an oil, it can relieve painful conditions such as neuralgia, sciatica, and rheumatism. Henbane reduces mucus secretions, as well as saliva and other digestive juices. Like its cousin deadly nightshade, it dilates the pupils. Hyoscine is commonly employed as a pre-operative anesthetic and in motion sickness formulations.

**Related Species** Other *Hyoscyamus* species are used in a similar way. The North African *H. muticus* is traditionally smoked by Bedouins to relieve toothache. See also deadly nightshade (*Atropa belladonna*, p. 69).

**⚠️ Cautions** Use only under professional supervision. Potentially toxic in overdose, henbane is subject to legal restrictions in some countries.

## *Hyssopus officinalis* (Lamiaceae) Hyssop

**Description** Semi-evergreen shrub growing to 2 ft (60 cm). Has narrow leaves and clusters of blue double-lipped flowers.

**Habitat & Cultivation** Native to southern Europe, hyssop grows freely in Mediterranean countries, especially in the Balkans and Turkey.



*Hyssop* has a positive effect on respiratory infections.

It prefers sunny, dry sites and is a common garden herb. The flowering tops are harvested when the plant is in flower in summer.

**Parts Used** Flowering tops, essential oil.

**Constituents** Contains terpenes (including marubiin, a diterpene), a volatile oil (consisting mainly of camphor, pinocamphone, and beta-pinene), flavonoids, tannins, and resin. Marubiin is a strong expectorant. Pinocamphone is toxic, and the volatile oil can cause epileptic seizures.

**History & Folklore** In the past, hyssop was so highly esteemed it was regarded as a virtual cure-all. An old saying went, "Whoever rivals hyssop's virtues, knows too much." In the 1st century CE, Dioscorides recommended a recipe containing a mixture of hyssop, figs (*Ficus carica*, p. 211), rue (*Ruta graveolens*, p. 265), honey, and water for treating a number of conditions, including pleurisy, asthma, tight-chestedness, respiratory congestion, and chronic coughs.

**Medicinal Actions & Uses** Currently an undervalued medicinal herb, hyssop is potentially useful as it is both calming and tonic. It has a positive effect when used to treat bronchitis and respiratory infections, especially where there is excessive mucus production. Hyssop appears to encourage the production of a more liquid mucus, and at the same time gently stimulates expectoration. This combined action clears thick and congested phlegm. Hyssop can irritate the mucous membranes, so it is best given after an infection has peaked, when the herb's tonic action encourages a general recovery. As a sedative, hyssop is a useful remedy against asthma in both children and adults, especially where the condition is exacerbated by mucus congestion. Like many herbs with a strong volatile oil, it soothes the digestive tract and can be an effective remedy against indigestion, gas, bloating, and colic.

**Cautions** Hyssop essential oil can induce epileptic seizures. It should only be used under professional supervision. Hyssop essential oil is subject to legal restrictions in some countries.

## *Ilex paraguariensis* syn. *I. paraguensis* (Aquifoliaceae) Maté

**Description** Evergreen shrub or small tree growing to 20 ft (6 m). Has large leaves, white flowers, and small reddish fruit.

**Habitat & Cultivation** Maté grows wild in northern Argentina, Paraguay, Uruguay, and southern Brazil, and is widely cultivated in Argentina, Spain, and Portugal. The leaves are picked when the berries are ripe, heated over a

wood fire, ground, and then stored in sacks for a year before being sold.

**Parts Used** Leaves.

**Constituents** Maté contains xanthine derivatives, including about 1.5% caffeine, about 0.2% theobromine, theophylline, and up to 16% tannins. The high tannin content means that maté should not be consumed with meals, as tannins impair the absorption of nutrients.

**Medicinal Actions & Uses** Maté is a



*Maté* makes a pleasant-tasting tea that has a stimulant, mildly analgesic, and diuretic effect.

traditional South American tea that increases short-term physical and mental energy levels. It is taken as a fortifying beverage in much the same way as tea (*Camellia sinensis*, p. 181) is consumed throughout Asia and Europe. Maté has properties similar to those of tea and coffee (*Coffea arabica*, p. 192). It stimulates the nervous system and is mildly analgesic and diuretic. As a medicinal herb, maté is used to treat headaches, migraine, neuralgic and rheumatic pain, fatigue, and mild depression. It has also been used in the treatment of diabetes.

**Related Species** *I. guayusa*, from Ecuador, is used in much the same way as maté but is also employed medicinally to treat malaria, liver pain, and syphilis. It is thought to aid digestion and cleanse the digestive tract.

## *Illicium verum* (Schisandraceae) Star Anise, Ba Jiao Hui Xian (Chinese)

**Description** Evergreen tree growing to 59 ft (18 m). Has tapering leaves, pale greenish-yellow flowers and star-shaped segmented fruit (seed pods).

**Habitat & Cultivation** Native to China, India and Vietnam, star anise grows in tropical regions across the world. The fruit is harvested when ripe.

**Parts Used** Fruit and seeds.